Customer Behavior & Risk Analysis

Business Objectives

- Analyze transaction behavior
- Segment customers
- Detect risk patterns

Data Cleaning & Preparation

Key Steps

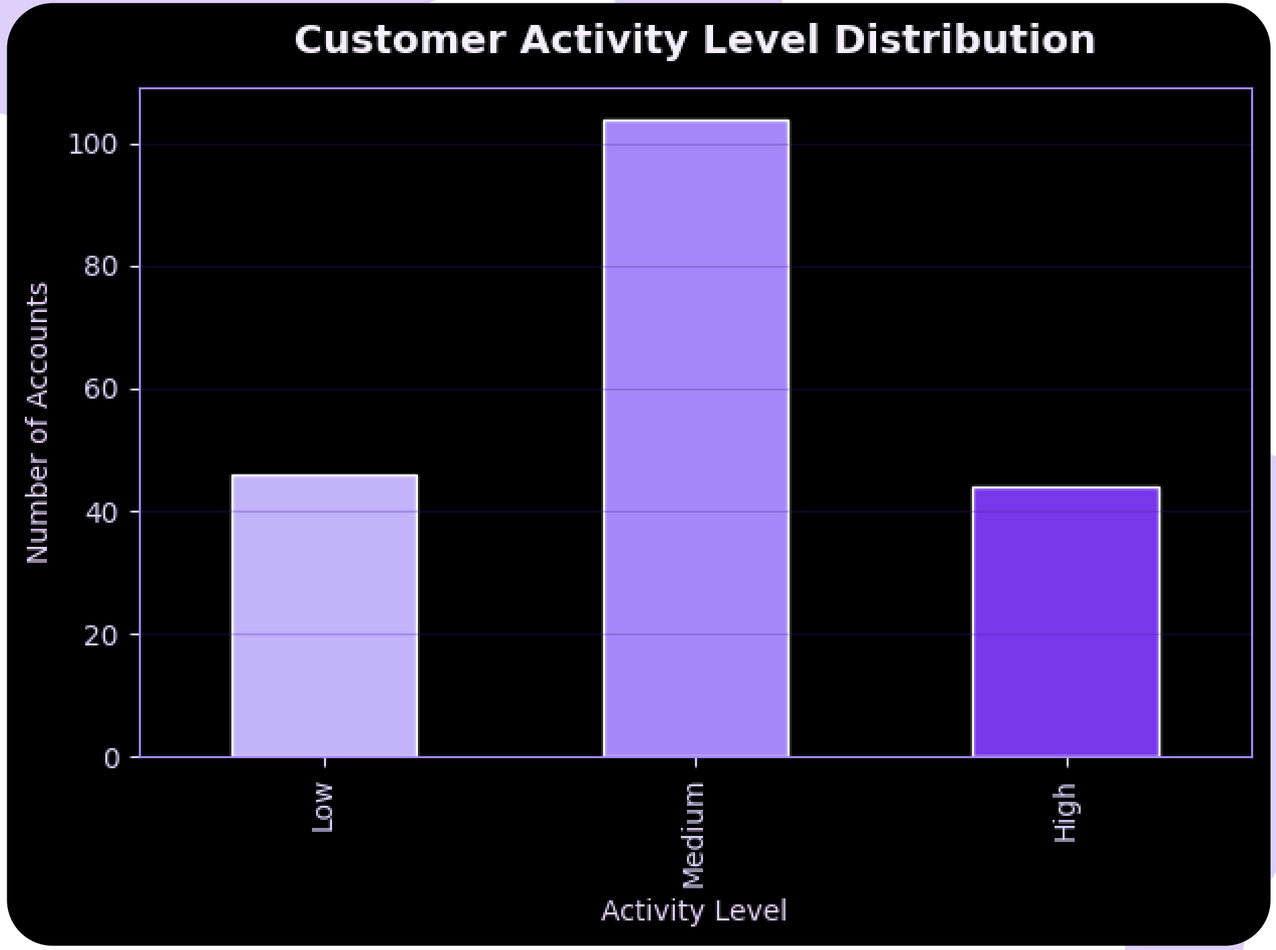
- Formatted dates & removed invalid entries
- Converted financial fields to numeric
- Standardized transaction & account types
- Handled missing values
- Created gap_days for activity analysis

Customer Behavior Insights

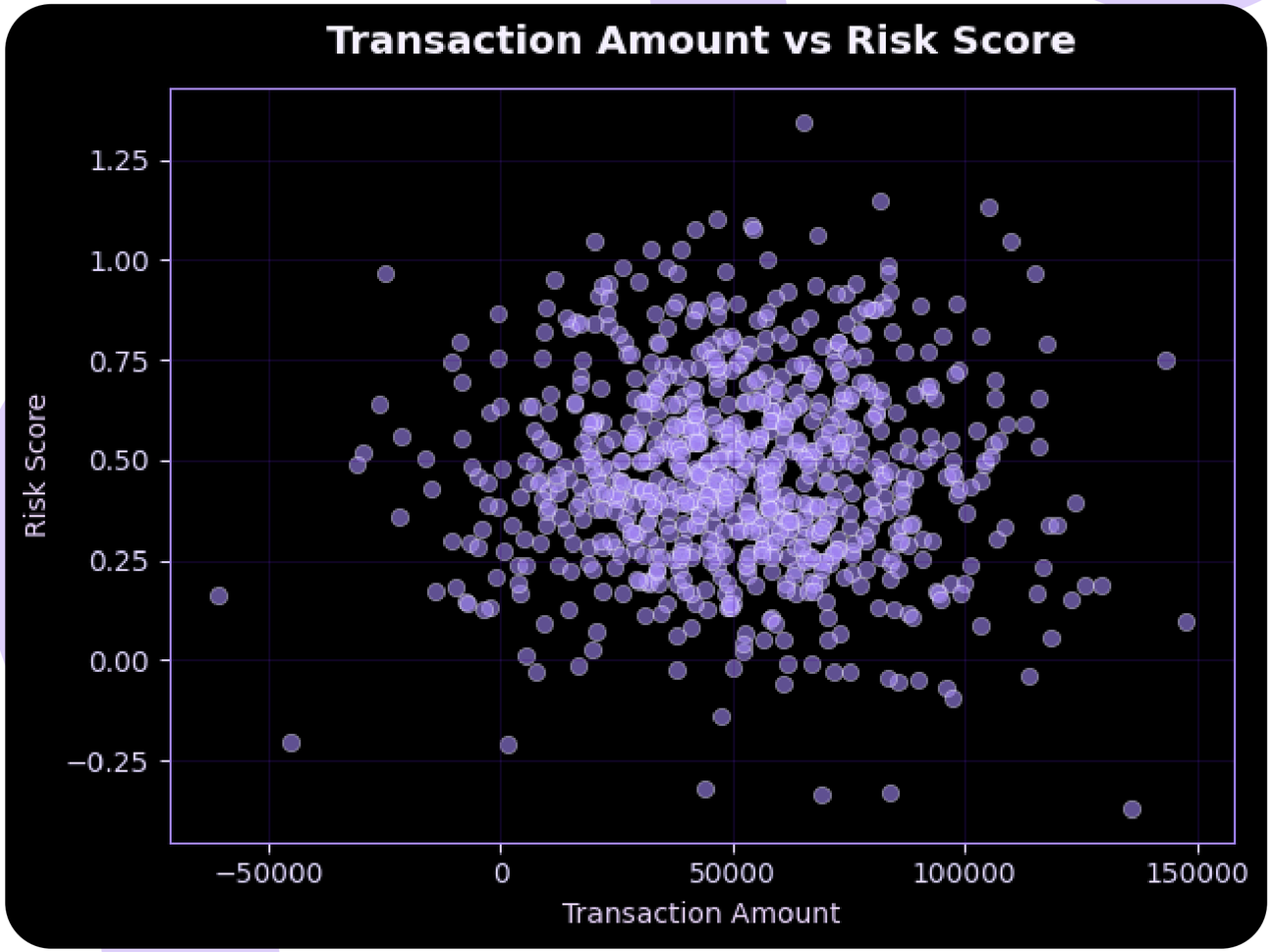
Key Insights

- **~60–70%** accounts show low activity
- **Top 25%** accounts drive major transaction volume
- Transactions show **high monthly variability**

- **Activity is concentrated in mid-level users**
- **Only a small segment is highly active**
- **Engagement is uneven across customers**



Risk & Anomaly Insights



Financial Risk Indicators

- **Top 15%** transactions flagged as large withdrawals
- High volatility accounts show **unstable balance patterns**
- Anomalies detected using statistical methods (Z-score)

23.7% Accounts Flagged as Suspicious

~1 in 4 Accounts Show Risk Signals (23.7%)

Recommendations

Engagement

60–70% Low Activity Users

→ Target with personalized campaigns

High Value

Top 25% Drive Volume

→ Focus on retention strategies

Risk Monitoring

23.7% Accounts at Risk

→ Enable proactive monitoring

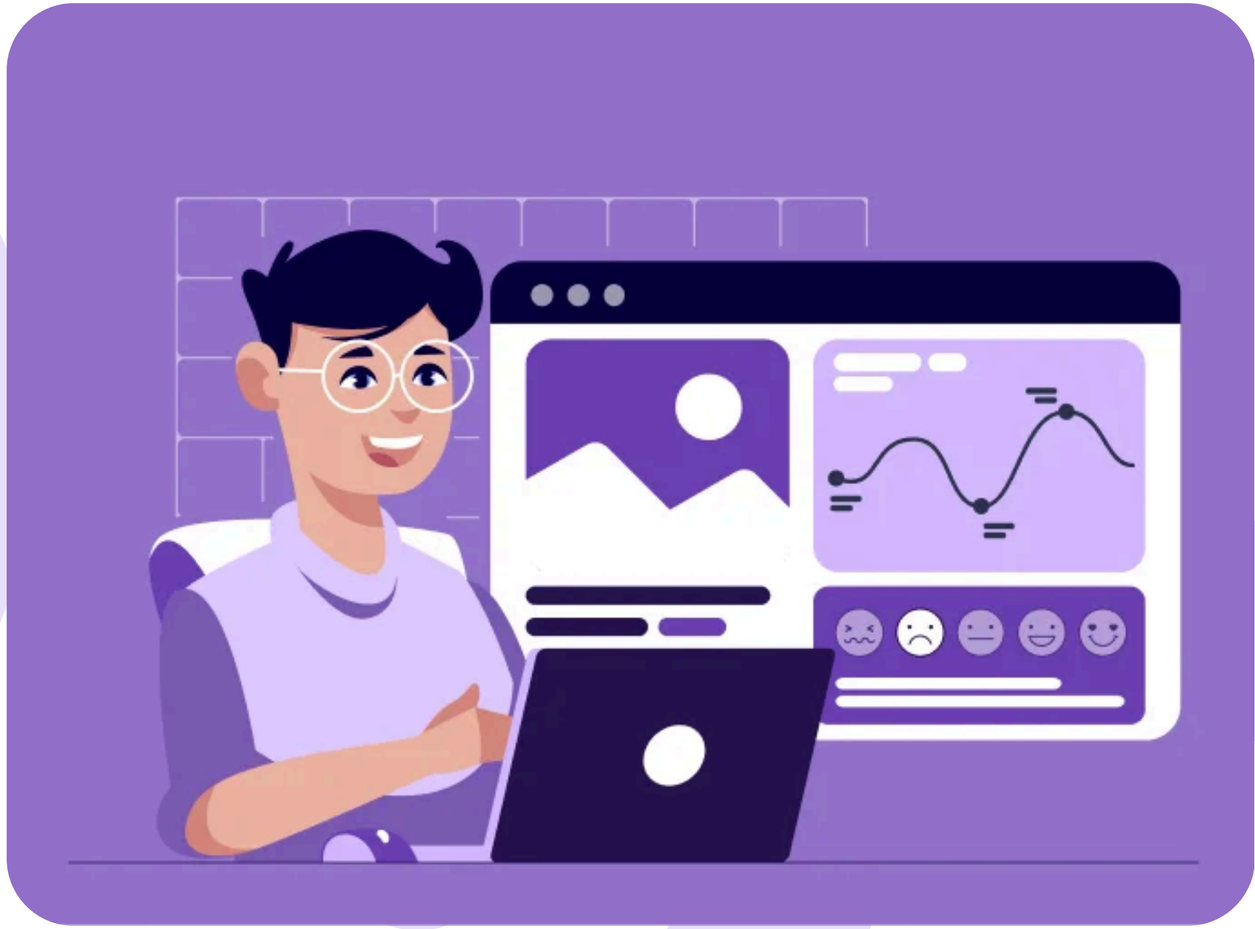
Anomaly Detection

Z-score Based Detection

→ Implement real-time alerts

Thank You

This analysis demonstrates how data-driven insights can uncover customer behavior patterns, identify financial risks, and support better decision-making

https://drive.google.com/file/d/1RWN0fa6bKvW-1an67XRB0tWEtYZ0-0f1/view?usp=sharing

Crafted and Designed by Manas